

G3_Task

Hello,

Please read the following carefully:

As a Software Engineering research team at the Polytechnique Montréal (Canada), we are interested in understanding developers' assessment of coding task difficulty. You will be provided with a series of 12 coding questions. Questions are divided into two parts:

- First, you will be given three instructions representing coding tasks and you will be asked to write, in a few words, what is the underlying task represented by all these instructions. The instructions represent the same tasks, but they vary in wording and information they give. In a nutshell, you have to distill the essence of the task and write it in a few words.

- In a second time, you will be asked to assess how difficult implementing this **task** would be. Note we ask the assessment based on the task you will have worded in the previous part, not based on any of the instructions presented earlier. First, you will be asked your subjective judgment on a scale from Very Easy to Very Difficult. Then, to make judging the difficulty more relatable, we adopt a similar approach as what is used in agile development and planning poker, i.e. you will be asked to evaluate how long it would take to implement a task (in minutes here). Consider in your assessment the knowledge needed to implement the task, the external information you would need to use, whether you would need the assistance of a LLM such as ChatGPT (and so the time it would take to debug the obtained code) etc. Finally, you will have to explain in a few words why you gave such difficulty ratings.

To give you an idea of the range of tasks you will need to assess, the first section of this questionnaire will be two examples we devised to illustrate the questionnaire.

Completing the task will take about 30 minutes.

Note also that all examples are written using Python 3.10 syntax.

Please make sure to answer all the questions to the best of your ability.

Please note that your identity and personal data will not be disclosed, while we plan to use the aggregated results and anonymized responses as part of a scientific publication. If you have any questions about the questionnaire or our research, please do not hesitate to contact us.

By proceeding, you consent to take part in the study under the conditions described previously.

Thank you,

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* Indique une question obligatoire

1. Please enter your prolific ID : *

Example of tasks

As we ask of you some estimation of difficulty and the tasks dealt in this questionnaire can be a bit different from what you can be used to as a developer, we give you examples of some tasks along with some assessments.

Note that this assessment is **our assessment**. You may have a different opinion based on your background. The idea of those examples is to give an extreme range of what type of tasks you can expect in the questionnaire.

As a reminder: You will first be given three instructions and asked to explain what the task is about in a few words. Then, you will have to rate the difficulty of the **task** you just described (not accounting for particular instruction). Rating the difficulty is done in two steps: subjective difficulty and approximate time needed. Finally, you will be asked to say why you gave such rating.

Example 1

Instruction 1

```
1 def subtract(c1, c2):  
2     """  
3     Deliver the result of subtracting two complex numbers,  
4     specified as 'c1' and 'c2', as a complex output.  
5     :param c1: The first complex number, complex.  
6     :param c2: The second complex number, complex.  
7     :return: The difference of the two complex numbers, complex.  
8     """
```

Instruction 2

```
1 def subtract(c1, c2):
2     """
3     The method receives two complex numbers 'c1' and 'c2', from which it
4     subtracts 'c2.real' from 'c1.real' to determine the real portion, and
5     subtracts 'c2.imag' from 'c1.imag' to determine the imaginary portion.
6     It constructs and returns a new complex number using these results
7     with the 'complex()' function.
8     :param c1: The first complex number, complex.
9     :param c2: The second complex number, complex.
10    :return: The difference of the two complex numbers, complex.
11    """
```

Instruction 3

```
1 def subtract(c1, c2):
2     """
3     Subtracts two complex numbers.
4     :param c1: The first complex number, complex.
5     :param c2: The second complex number, complex.
6     :return: The difference of the two complex numbers, complex.
7     >>> complexCalculator = ComplexCalculator()
8     >>> complexCalculator.subtract(1+2j, 3+4j)
9     (-2-2j)
10    """
```

Answer

The task encapsulated by the **subtract**

function

is to compute the subtraction between two complex numbers.

This task has a subjective difficulty of **Very Easy** as assessed by us and would take about **1 min**

to implement. There is no particular difficulty, the task is very straightforward without any corner case, it just requires to know what a complex is, and the standard Python library for it.

Example 2

Context

```
1 import numpy as np
2 class MetricsCalculator2:
3     """
4     The class provides to calculate Mean Reciprocal Rank (MRR) and Mean Average Precision (MAP)
5     based on input data, where MRR measures the ranking quality and MAP measures the average precision.
6     """
7
8     def __init__(self):
9         pass
10
11     @staticmethod
12     def map(data):
13         pass
```

Instruction 1

```
1 def mrr(data):
2     """
3     Compute the MRR of the input data. MRR is a widely used evaluation index.
4     It is the mean of reciprocal rank.
5     :param data: the data must be a tuple, list 0,1, eg. ([1,0,...],5).
6     In each tuple (actual result,ground truth num),ground truth num is the total ground num.
7     ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5),([1,0,...],6),([0,0,...],5)].
8     1 stands for a correct answer, 0 stands for a wrong answer.
9     :return: if input data is list, return the recall of this list. if the input data is list of list, return the
10     average recall on all list. The second return value is a list of precision for each input.
11     >>> MetricsCalculator2.mrr([([1, 0, 1, 0], 4)])
12     >>> MetricsCalculator2.mrr([([1, 0, 1, 0], 4), ([0, 1, 0, 1], 4)])
13     1.0, [1.0]
14     0.75, [1.0, 0.5]
15
16     """
```

Instruction 2

```
1 def mrr(data):
2     """
3     Assess the Mean Reciprocal Rank (MRR) from the input 'data'.
4     MRR assesses the average of reciprocal ranks of outcomes.
5     The input structure 'data' must be a tuple consisting of a list of
6     binary values and its total count or a list of such tuples.
7     Each binary value (0 or 1) represents the correctness of an answer.
8     If 'data' is a tuple, the function returns its mean reciprocal rank.
9     If 'data' is a list, the function returns the overall average MRR along
10     with a list showing reciprocal ranks for each component tuple.
11
12     :param data: the data must be a tuple, list 0,1, eg. ([1,0,...],5).
13     In each tuple (actual result,ground truth num),ground truth num is the total ground num.
14     ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5),([1,0,...],6),([0,0,...],5)].
15     1 stands for a correct answer, 0 stands for a wrong answer.
16     :return: if input data is list, return the recall of this list. if the input data is list of list, return the
17     average recall on all list. The second return value is a list of precision for each input.
18
19     """
```

Instruction 3

```
1 def mrr(data):
2     """
3     Evaluate the Mean Reciprocal Rank (MRR) from 'data'. This metric calculates the mean of reciprocal
4     ranks for correct answers. Input 'data' could be either a tuple containing binary correctness indicators
5     and the aggregate count of answers, or multiple tuples of this type in a list. For each tuple, the function
6     measures reciprocal ranks by multiplying the binary indicators by reciprocals of their positions, finding the
7     earliest nonzero value to establish the rank of the first correct response. If the input 'data' is a tuple,
8     the MRR calculated for this specific tuple is returned; if it's a list, the function summarizes by providing
9     both the average MRR and a list of individual MRR values for each tuple.
10
11     :param data: the data must be a tuple, list 0,1, eg. ([1,0,...],5).
12     In each tuple (actual result, ground truth num), ground truth num is the total ground num.
13     ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5), ([1,0,...],6), ([0,0,...],5)].
14     1 stands for a correct answer, 0 stands for a wrong answer.
15     :return: if input data is list, return the recall of this list. if the input data is list of list, return the
16     average recall on all list. The second return value is a list of precision for each input.
17
18     """
```

Answer

The task encapsulated by the **mrr** function is to compute the Mean Reciprocal Rank of binary tuples data.

This task has a subjective difficulty of **Very Hard** as assessed by us and would take about **40 min**

to implement. The difficulty mainly branches from dealing with the tuple data and how the formula is processed accordingly (with a few corner cases). This can take additional time if the Mean Reciprocal Rank is not known.

Question 1:

Below are instructions for implementing the function.

Instruction 1

```
1 def incr_list(l: list):
2     """
3     Write a function named "incr_list" which takes a list "l"
4     as input and returns a new list. The function iterates over
5     each integer "x" in the input list "l" and increments each
6     by 1 using the expression "x + 1". These incremented values
7     are collected into a new list using a list comprehension.
8     The resulting list, which contains each original element
9     incremented by 1, is then returned.
10    """
```

Instruction 2

```
1 def incr_list(l: list):
2     """Return list with elements incremented by 1.
3     >>> incr_list([1, 2, 3])
4     [2, 3, 4]
5     >>> incr_list([5, 3, 5, 2, 3, 3, 9, 0, 123])
6     [6, 4, 6, 3, 4, 4, 10, 1, 124]
7     """
```

Instruction 3

```
1 def incr_list(l: list):
2     """
3     Formulate a function 'incr_list' that inputs a list of integers
4     and returns another list where each element is raised by 1.
5     """
```

2. In a few words, what is the task this function aims to implement? *

3. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

4. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

5. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 2:

Below are instructions for implementing the function:

Context

```
1 class NumericEntityUnescaper:
2     """
3     This is a class that provides functionality to replace numeric
4     entities with their corresponding characters in a given string.
5     """
6
7     def __init__(self):
8         pass
9
10    def replace(self, string):
11        pass
12
13
14    @staticmethod
15    def is_hex_char(char):
16
17
18
19
```

Instruction 1

```
1 def is_hex_char(char):
2     """
3     Determines whether a given character "char" is a hexadecimal digit.
4     Returns "True" if "char" is a hexadecimal digit, otherwise "False".
5     The function checks if the character is either a digit by invoking
6     Python's inbuilt "isdigit()" method or a letter from 'a' to 'f'
7     (considering both lowercase and uppercase), by converting "char" to
8     lowercase using "lower()" method and then checking the range using Python's
9     comparison operators.
10    :param char: str, the character to check.
11    :return: bool, True if the character is a hexadecimal digit, False otherwise.
12    """
```

Instruction 2

```
1 def is_hex_char(char):
2     """
3     Determines whether a given character is a hexadecimal digit.
4     :param char: str, the character to check.
5     :return: bool, True if the character is a hexadecimal digit, False otherwise.
6     >>> NumericEntityUnescaper.is_hex_char('a')
7     True
8     """
```


Instruction 3

```
1 def is_hex_char(char):  
2     """  
3     Determine if 'char' is considered a hexadecimal digit, returning 'True'  
4     for valid hex characters and 'False' otherwise. The function verifies  
5     if the character is a digit using 'isdigit()', or falls between 'a' and 'f'  
6     after conversion to lowercase, handling both cases by employing standard  
7     Python methods and comparison.  
8     :param char: str, the character to check.  
9     :return: bool, True if the character is a hexadecimal digit, False otherwise.  
10    """
```

6. In a few words, what is the task this function aims to implement? *

7. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

8. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

9. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 3:

Below are instructions for implementing the function:

Context

```
1 def encode_shift(s: str):  
2     """  
3     returns encoded string by shifting every character by 5 in the alphabet.  
4     """  
5     return "".join([chr(((ord(ch) + 5 - ord("a")) % 26) + ord("a")) for ch in s])
```

Instruction 1

```
1 def decode_shift(s: str):
2     """
3     Implement a function titled 'decode_shift' which takes an encoded
4     string 's' and decodes it. Each character in 's' was initially shifted
5     5 positions forward in the alphabet. This function will shift each
6     character in 's' backward by 5 positions using
7     the formula 'chr((ord(ch) - ord('a') - 5 + 26) % 26 + ord('a'))'
8     for each character 'ch' in 's', and it will account for
9     looping back through the alphabet as necessary.
10    """
```

Instruction 2

```
1 def decode_shift(s: str):
2     """
3     takes as input string encoded with encode_shift function. Returns decoded string.
4     """
```

Instruction 3

```
1 def decode_shift(s: str):
2     """
3     Craft a function 'decode_shift' which receives a string that
4     has each character shifted forwards by 5 positions in the
5     alphabet (with wrap-around). The function should return a string
6     where each character is shifted back by 5 positions to decode it,
7     considering wrap-around when necessary.
8     """
```

10. In a few words, what is the task this function aims to implement? *

11. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

12. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

13. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 4:

Below are instructions for implementing the function:

Context

```
1 import sqlite3
2 import pandas as pd
3 class DatabaseProcessor:
4     """
5     This is a class for processing a database, supporting to create tables,
6     insert data into the database, search for data based on name, and delete
7     data from the database.
8     """
9
10    def __init__(self, database_name):
11        """
12        Initialize database name of database processor
13        """
14        self.database_name = database_name
15
16
17    def create_table(self, table_name, key1, key2):
18        pass
19
20    def search_database(self, table_name, name):
21        pass
22
23    def delete_from_database(self, table_name, name):
24        pass
25
26
27
```

Instruction 1

```
1 def insert_into_database(self, table_name, data):
2     """
3     Connect to the database and prepare a cursor for executing
4     SQL operations. Traverse through each dictionary in the list 'data',
5     with each representing a row in the database where the keys are
6     column names and the values are the respective row data. Generate
7     and execute an SQL insert query for each dictionary to incorporate
8     the data into the specified 'table_name'. Conclude by committing the
9     transaction and closing the connection to uphold data integrity.
10    :param table_name: str, the name of the table to insert data into.
11    :param data: list, a list of dictionaries where each dictionary
12                 represents a row of data.
13    """
```

Instruction 2

```
1 def insert_into_database(self, table_name, data):
2     """
3     Insert data into the specified table in the database.
4     :param table_name: str, the name of the table to insert data into.
5     :param data: list, a list of dictionaries where each dictionary
6     represents a row of data.
7     >>> db.insert_into_database('user', [
8     {'name': 'John', 'age': 25},
9     {'name': 'Alice', 'age': 30}
10    ])
11    """
```

Instruction 3

```
1 def insert_into_database(self, table_name, data):
2     """
3     Insert data into the specified table "table_name" in the database.
4     The data to be inserted is provided as a list of dictionaries "data",
5     where each dictionary represents a row with keys as column names and
6     values as data entries. Initiate a connection to the database and
7     create a cursor to execute queries. Loop through the list "data",
8     and for each dictionary in the list, generate an SQL insert query
9     to add the data into the table. Close the database connection after
10    committing the changes to ensure data integrity.
11    :param table_name: str, the name of the table to insert data into.
12    :param data: list, a list of dictionaries where each dictionary
13    represents a row of data.
14    """
```

14. In a few words, what is the task this function aims to implement? *

15. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

16. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

17. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 5:

Below are instructions for implementing the function:

Instruction 1

```
1 def is_happy(s):
2     """
3     Define a function 'is_happy' that accepts a string parameter 's'.
4     The function first checks if 's' is under three characters long,
5     and returns False in such cases. Then, it proceeds to cycle through
6     the string, testing every trio of consecutive characters
7     from 's[i]' to 's[i + 2]'.
8     It assesses whether 's[i]', 's[i + 1]', and 's[i + 2]' are different
9     from each other. Any repetition among these characters results in a
10    False return. If no identical trios are detected, the function deems
11    the string as happy and returns True.
12    """
```

Instruction 2

```
1 def is_happy(s):
2     """
3     You are given a string s.
4     Your task is to check if the string is happy or not.
5     A string is happy if its length is at least 3 and
6     every 3 consecutive letters are distinct
7     For example:
8     is_happy(a) => False
9     is_happy(aa) => False
10    is_happy(abcd) => True
11    is_happy(aabb) => False
12    is_happy(adb) => True
13    is_happy(xyy) => False
14    """
```

Instruction 3

```
1 def is_happy(s):
2     """
3     Construct a function 'is_happy' to ascertain if a string can be considered 'happy'.
4     Initially, the function checks if the string is shorter than three characters,
5     in which case it is automatically labelled as not happy. For strings that meet
6     the length requirement, the function examines every group of three consecutive
7     characters to ensure they are all different. Any group with duplicate characters
8     renders the string not happy. Only if every group has unique characters does the
9     string qualify as happy.
10    """
```

18. In a few words, what is the task this function aims to implement? *

19. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

20. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

21. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 6:

Below are instructions for implementing the function:

Context

```
1 class IPAddress:
2     """
3     This is a class to process IP Address, including validating,
4     getting the octets and obtaining the binary representation of
5     a valid IP address.
6     """
7
8     def __init__(self, ip_address):
9         """
10        Initialize the IP address to the specified address
11        :param ip_address:string
12        """
13        self.ip_address = ip_address
14
15
16    def get_octets(self):
17        pass
18
19    def get_binary(self):
20        pass
21
22
23
```

Instruction 1

```
1 def is_valid(self):
2     """
3     Examine whether the 'self.ip_address' comprises a valid IP address,
4     which should include four decimal octets separated by dots ('.').
5     Every octet needs to be a number between 0 and 255. Initially,
6     'is_valid' uses the 'split' method on 'self.ip_address' with '.'
7     as the split character, storing the split parts in 'octets'.
8     The function checks that 'octets' has exactly four segments.
9     Each segment in 'octets' is evaluated to see if it qualifies as
10    a digit and lies within the allowed 0 to 255 range, using 'isdigit'
11    and integer validation. Return 'True' if every condition is fulfilled,
12    indicating a valid IP address; else, 'False'.
13    :return: bool
14    """
```

Instruction 2

```
1 def is_valid(self):
2     """
3     Judge whether the IP address is valid, that is, whether the IP
4     address is composed of four Decimal digits separated by '.'.
5     Each digit is greater than or equal to 0 and less than or equal
6     to 255
7     :return: bool
8     >>> ipaddress = IPAddress("10.10.10.10")
9     >>> ipaddress.is_valid()
10    True
11    """
```

Instruction 3

```
1 def is_valid(self):
2     """
3     Check if an IP address is correctly formatted with four dot-separated
4     decimal octets, where each octet is a number between 0 and 255.
5     The function should output 'True' if the IP address conforms to this format,
6     otherwise 'False'.
7     :return: bool
8     """
```

22. In a few words, what is the task this function aims to implement? *

23. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

24. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

25. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 7:

Below are instructions for implementing the function:

Instruction 1

```
1 def greatest_common_divisor(a: int, b: int) -> int:
2     """
3     Write 'greatest_common_divisor' function that takes
4     two integer inputs. It aims to determine and return
5     the greatest common divisor (GCD) using the Euclidean
6     method. An encapsulated function recurses with these
7     integers, alternates their order and reduces them by
8     modulus until one becomes zero, at which it's concluded
9     as the GCD by the initial function.
10    """
```

Instruction 2

```
1 def greatest_common_divisor(a: int, b: int) -> int:
2     """ Return a greatest common divisor of two integers a and b
3     >>> greatest_common_divisor(3, 5)
4     1
5     >>> greatest_common_divisor(25, 15)
6     5
7     """
```

Instruction 3

```
1 def greatest_common_divisor(a: int, b: int) -> int:
2     """
3     Draft a function 'greatest_common_divisor' that accepts
4     two integers 'a' and 'b' as inputs. This function is configured
5     to ascertain and provide the greatest common divisor of the
6     input integers using the Euclidean formula. Contained within
7     this function is a recursive auxiliary function, 'query_gcd',
8     which receives 'a' and 'b'. It pushes the recursion forward by
9     self-invoking with 'b' and 'a % b' (where 'a % b' represents 'a' modulo 'b')
10    until 'b' becomes null, at which point it returns 'a' as the divisor. Initially,
11    'greatest_common_divisor' activates 'query_gcd' with 'a' and 'b', concluding
12    by returning the computed GCD.
13    """
```

26. In a few words, what is the task this function aims to implement? *

27. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

28. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

29. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 8:

Below are instructions for implementing the function:

Context

```
1 class JobMarketplace:
2     """
3     This is a class that provides functionalities to publish positions,
4     remove positions, submit resumes, withdraw resumes, search for positions,
5     and obtain candidate information.
6     """
7
8     def __init__(self):
9         self.job_listings = []
10        self.resumes = []
11
12    def remove_job(self, job):
13        pass
14
15    def submit_resume(self, name, skills, experience):
16        pass
17
18    def withdraw_resume(self, resume):
19        pass
20
21    def search_jobs(self, criteria):
22        pass
23
24    def get_job_applicants(self, job):
25        pass
26
27
28
```

Instruction 1

```
1 def post_job(self, job_title, company, requirements):
2     """
3     Include a job posting to the 'job_listings' list through a
4     function that accepts 'job_title', 'company', and 'requirements'.
5     Commence by forming a dictionary with keys that reflect the job title,
6     company name, and the array of job requirements. Append this dictionary
7     to the 'job_listings' list to effectively populate the marketplace
8     with the job.
9     :param job_title: The title of the position,str.
10    :param company: The company of the position,str.
11    :param requirements: The requirements of the position,list.
12    :return: None
13    """
```

Instruction 2

```
1 def post_job(self, job_title, company, requirements):
2     """
3     This function is used to publish positions, and add the position
4     information to the job_listings list.
5     :param job_title: The title of the position, str.
6     :param company: The company of the position, str.
7     :param requirements: The requirements of the position, list.
8     :return: None
9
10    >>> jobMarketplace = JobMarketplace()
11    >>> jobMarketplace.post_job("""Software Engineer""",
12    ""ABC Company"", ['requirement1', 'requirement2'])
13    >>> jobMarketplace.job_listings
14    [{'job_title': 'Software Engineer', 'company':
15     'ABC Company', 'requirements': ['requirement1', 'requirement2']}]
```

Instruction 3

```
1 def post_job(self, job_title, company, requirements):
2     """
3     Publish a job position by adding the position information to
4     the "job_listings" list. The function should accept three
5     parameters: "job_title", "company", and "requirements".
6     Begin by constructing a dictionary representing the job with
7     keys corresponding to the job title, company name, and list of
8     job requirements. This dictionary is then appended to the
9     "job_listings" list to effectively add the job to the marketplace.
10    :param job_title: The title of the position, str.
11    :param company: The company of the position, str.
12    :param requirements: The requirements of the position, list.
13    :return: None
14    """
```

30. In a few words, what is the task this function aims to implement? *

31. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

32. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

33. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 9:

Below are instructions for implementing the function:

Instruction 1

```
1 def odd_count(lst):
2     """
3     Develop a function 'odd_count' which accepts a list called
4     'lst' filled with strings, with each string only containing
5     numeric characters. This function should generate a new list
6     'ans' with each item being a formatted statement: 'the number
7     of odd elements in the string i of the input.', substituting
8     all the letters 'i' with the count of odd digits in the respective
9     string from 'lst'. Navigate through each string 's' in 'lst', use a
10    lambda in a filter function to compute the count of odd digits,
11    and place this count in 'odd_cnt'. Subsequently, replace all the 'i'
12    in the template with 'odd_cnt' and append each completed message to 'ans'.
13    Return 'ans' at the conclusion.
14    """
```

Instruction 2

```
1 def odd_count(lst):
2     """
3     Given a list of strings, where each string consists of only digits, return a list.
4     Each element i of the output should be "the number of odd elements in the
5     string i of the input." where all the i's should be replaced by the number
6     of odd digits in the i'th string of the input.
7
8     >>> odd_count(['1234567'])
9     ["the number of odd elements 4n the str4ng 4 of the 4nput."]
10    >>> odd_count(['3','1111111'])
11    ["the number of odd elements 1n the string 1 of the 1nput.",
12     "the number of odd elements 8n the str8ng 8 of the 8nput."]
13    """
```

Instruction 3

```
1 def odd_count(lst):
2     """
3     Formulate a function 'odd_count' which deals with an input
4     list of numeric strings. Output should be a list where each
5     entry is formatted as 'the number of odd elements in the string
6     i of the input.', replacing all the letters 'i' with the count
7     of odd digits in each particular string.
8     """
```

34. In a few words, what is the task this function aims to implement? *

35. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

36. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

37. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 10:

Below are instructions for implementing the function:

Context

```
1 from collections import Counter
2 import re
3 class NLPDataProcessor2:
4     """
5     The class processes NLP data by extracting words from a list of strings,
6     calculating the frequency of each word, and returning the top 5 most frequent words.
7     """
8
9     def process_data(self, string_list):
10         pass
11
12     def process(self, string_list):
13         pass
14
15
16
```

Instruction 1

```
1 def calculate_word_frequency(self, words_list):
2     """
3     Compute the frequency of each word from the 'words_list' array list,
4     sort these frequencies in a descending order. Initially, it establishes a
5     'word_frequency' Counter. As it iterates through each individual sublist
6     contained in 'words_list', it enhances the 'word_frequency' using the 'update'
7     method of the Counter with the encountered words. After all sublists are processed,
8     the word frequencies are sorted in descending order using a lambda function in the
9     'sorted' function. The top five frequencies are then isolated, converted into a
10    dictionary labeled 'top_5_word_frequency', which consists of words and their respective
11    frequencies, and this is what the function returns.
12    :param words_list: a list of words lists
13    :return: top 5 word frequency dictionary, a dictionary of word frequency,
14            key is word, value is frequency
15    """
```

Instruction 2

```
1 def calculate_word_frequency(self, words_list):
2     """
3     Calculate the word frequency of each word in the list of words list,
4     and sort the word frequency dictionary by value in descending order.
5     :param words_list: a list of words lists
6     :return: top 5 word frequency dictionary, a dictionary of word frequency,
7     key is word, value is frequency
8     >>> NLPDataProcessor.calculate_word_frequency(['this', 'is', 'a', 'test'],
9     ['this', 'is', 'another', 'test'])
10    {'this': 2, 'is': 2, 'test': 2, 'a': 1, 'another': 1}
11    """
```

Instruction 3

```
1 def calculate_word_frequency(self, words_list):
2     """
3     Create a function to calculate how often each word appears in
4     'words_list' where the list contains nested lists of words,
5     sorting this information into a frequency dictionary descending by value.
6     The function first merges word counts across all sublists in 'words_list'.
7     Then, it sorts these totals in descending order and extracts the top five
8     words with the highest frequencies, returning them as a dictionary with
9     words as keys and their occurrence counts as values.
10    :param words_list: a list of words lists
11    :return: top 5 word frequency dictionary, a dictionary of word frequency,
12    key is word, value is frequency
13
14    """
```

38. In a few words, what is the task this function aims to implement? *

39. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

40. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

41. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 11:

Below are instructions for implementing the function

Instruction 1

```
1 def minSubArraySum(nums):  
2     """  
3     Create a function 'minSubArraySum' that calculates  
4     the smallest sum of a non-empty contiguous subset of  
5     an integer array.  
6     """
```

Instruction 2

```
1 def minSubArraySum(nums):  
2     """  
3     Given an array of integers nums, find the minimum sum of  
4     any non-empty sub-array of nums.  
5     Example  
6     minSubArraySum([2, 3, 4, 1, 2, 4]) == 1  
7     minSubArraySum([-1, -2, -3]) == -6  
8     """
```

Instruction 3

```
1 def minSubArraySum(nums):  
2     """  
3     Construct a function named 'minSubArraySum' that takes a list of  
4     integers 'nums' as argument. The function strives to compute and  
5     output the smallest sum from any sub-array within 'nums'.  
6     Initially, it assesses whether all figures in 'nums' are non-negative  
7     using 'all(x >= 0 for x in nums)'. If this is the case, the smallest  
8     integer from 'nums' is returned through 'min(nums)'. If there are any  
9     negative integers, the function initializes 's' and 'ans' as zero.  
10    For each integer 'x' in 'nums', it is added to 's' and 'ans' is  
11    updated to the lesser of the two values 'ans' and 's'. If 's' reaches  
12    or exceeds zero during the process, it is reset to zero to start over  
13    the calculation for a new sub-array sum. Finally, the function issues  
14    'ans', which is the lowest subarray sum found.  
15    """
```

42. In a few words, what is the task this function aims to implement? *

43. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

44. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

45. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 12:

Below are instructions for implementing the function:

Context

```
1 class BigNumCalculator:
2     """
3     This is a class that implements big number calculations,
4     including adding, subtracting and multiplying.
5     """
6
7     def add(num1, num2):
8         pass
9
10    @staticmethod
11    def multiply(num1, num2):
12        pass
13
14
15    @staticmethod
16    def subtract(num1, num2):
17
```

Instruction 1

```
1 def subtract(num1, num2):
2     """
3     Subtracts two big numbers.
4     :param num1: The first number to subtract,str.
5     :param num2: The second number to subtract,str.
6     :return: The difference of the two numbers,str.
7     >>> bigNum = BigNumCalculator()
8     >>> bigNum.subtract("12345678901234567890", "98765432109876543210")
9     '-86419753208641975320'
10    """
```

Instruction 2

```
1 def subtract(num1, num2):
2     """
3     Determine and output the subtraction result of
4     two strings 'num1' and 'num2', which represent
5     large numbers, returning the result as a string.
6     :param num1: The first number to subtract,str.
7     :param num2: The second number to subtract,str.
8     :return: The difference of the two numbers,str.
9     """
```

Instruction 3

```
1 def subtract(num1, num2):
2     """
3     Compute the difference by subtracting 'num2' from
4     'num1' – both represented as strings of arbitrary
5     length – returning the calculation as a string.
6     If 'num1' is the shorter or smaller number,
7     put 'num2' first and signal that the result will
8     be negative. Firstly, pad the numbers to equal length,
9     subtract sequentially from the rightmost digit accounting
10    for borrows, build this into a result string, strip leading
11    zeros, prepend '-' for negative outcomes, and finally,
12    return the subtracted result.
13    :param num1: The first number to subtract,str.
14    :param num2: The second number to subtract,str.
15    :return: The difference of the two numbers,str.
16
17    """
```

46. In a few words, what is the task this function aims to implement? *

47. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

48. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

49. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Demographics Information

50. What is your highest education qualification? *

Une seule réponse possible.

- ☐ Ph.D.
- ☐ Master's Degree
- ☐ Bachelor Degree
- ☐ Autre :

51. What is your field of work? (Engineer, Researcher, Developer, etc.) *

52. How many years of development experience do you have? *

Une seule réponse possible.

- ☐ Less than 5
- ☐ Between 5 and 10
- ☐ More than 10

53. Thank you for completing the survey! Any feedback you would like to provide?

Google Forms

